EC200D&ECx00G&EC600U Series

Firmware Download Guide

LTE Standard Module Series

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About the Document

Revision History

Version	Date	Author	Description
-	2023-04-06	Mahat XU	Creation of the document
1.0	2023-06-07	Mahat XU	First official release
1.1	2023-08-18	Mahat XU	Added applicable module EC600G series.
1.2	2024-07-16	Mahat XU	Deleted EG800G series according to Quectel's product marketing strategies.

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1 Introduction

This document mainly introduces the firmware download method of Quectel EC200D series, ECx00G family and EC600U series modules.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
-	EC200D Series
ECx00G	EC200G-CN
	EC600G Series
	EC800G-CN
-	EC600U Series

2 Firmware Download

2.1. Driver Installation

Before downloading the firmware, you need to install the device driver. Unzip the USB driver package and install it according to the installation instructions in the USB driver package.

NOTE

Please contact Quectel Technical Support (support@quectel.com) for the latest version of the driver.

2.2. Port Configuration

After the driver is installed, connect the module to the host through USB.

- If the module has never downloaded firmware before, the module will enter the download mode after the module's booting, and the port can be viewed in the device manager of the host, as shown in ***Figure 1***.

- If the module has downloaded the firmware, the ports can be viewed in the device manager of the host after the module's booting, as shown in ***Figure 2***, ***Figure 3*** and ***Figure 4***. And then you can do either of the following options:

  (1) Send **AT+QDOWNLOAD** to reboot the module and enter the download mode.
  (2) Short-circuit the USB_BOOT and VDD_EXT pins of the module, reboot the module and enter the forced download mode. Then the port can be viewed as the following figure. Noted that the USB_BOOT pin needs to be connected in series with a 4.7 kΩ resistor and then shorted to the VDD_EXT pin, with a TVS tube added for protection at the same time. For details, see the hardware design documents of each module.

Figure 1: Port Viewed When the Module Has Never Downloaded the Firmware

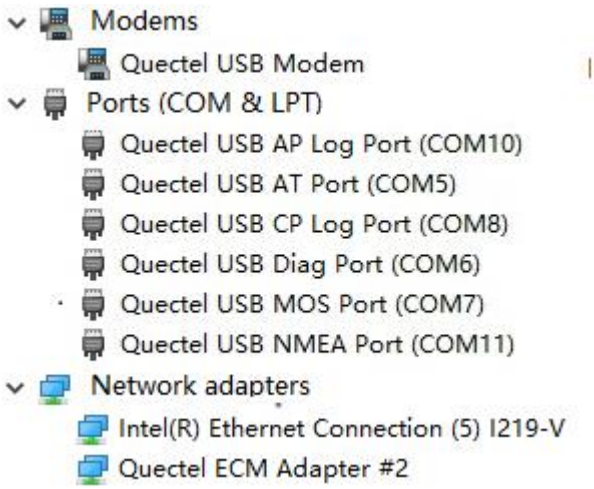

Figure 2: Ports Viewed When the Module Has Downloaded the Firmware (EC600U Series)

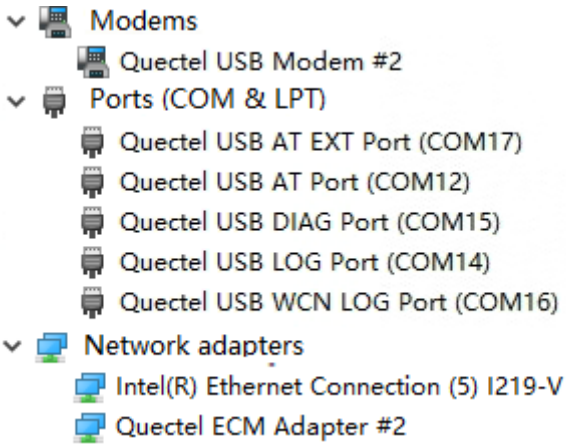

Figure 3: Ports Viewed When the Module Has Downloaded the Firmware (EC200D Series)

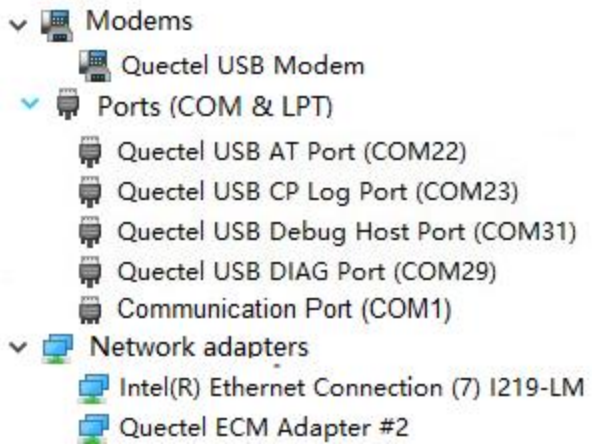

Figure 4: Ports Viewed When the Module Has Downloaded the Firmware (ECx00G)

2.3. Firmware Download

This document takes the firmware download tool *UpgradeDownload_R24.0. 0003.7z* as an example. After decompression, run the firmware download program *UpgradeDownload.exe* in the *Bin* directory and enter the download interface of the UpgradeDownload tool.

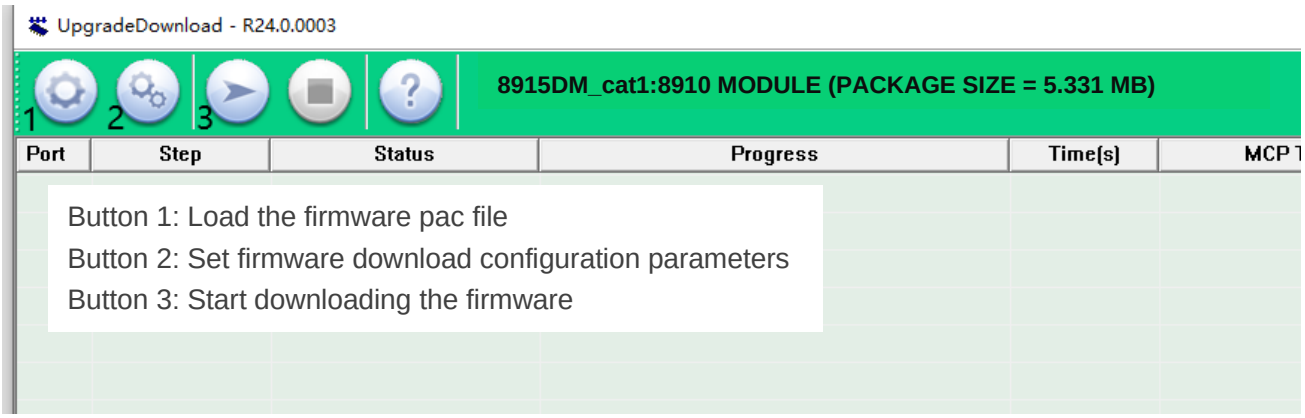

Figure 5: Firmware Download Tool: UpgradeDownload

NOTE

Please contact Quectel Technical Support (support@quectel.com) for the latest firmware download tool.

1. Click Button 2 in the main interface to configure parameters as needed, such as baud rate (only for physical serial ports).

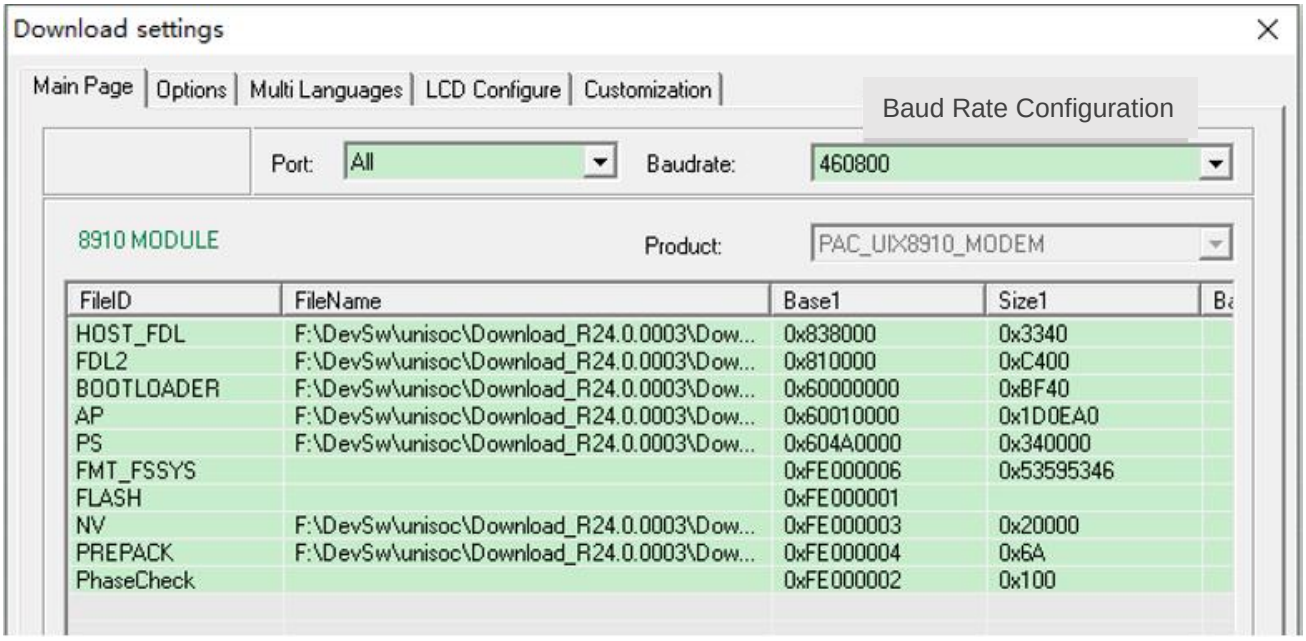

Figure 6: Parameter Configurations of UpgradeDownload

NOTE

EC200D series module does not support downloading the firmware through serial port, so you can directly proceed to the next step without configuring the baud rate.

Configure whether to reboot the module automatically after firmware download in the Options interface. If you need to reboot the module, choose "**Reset to Normal**" in "End of download" and click "**OK**" to complete the configuration.

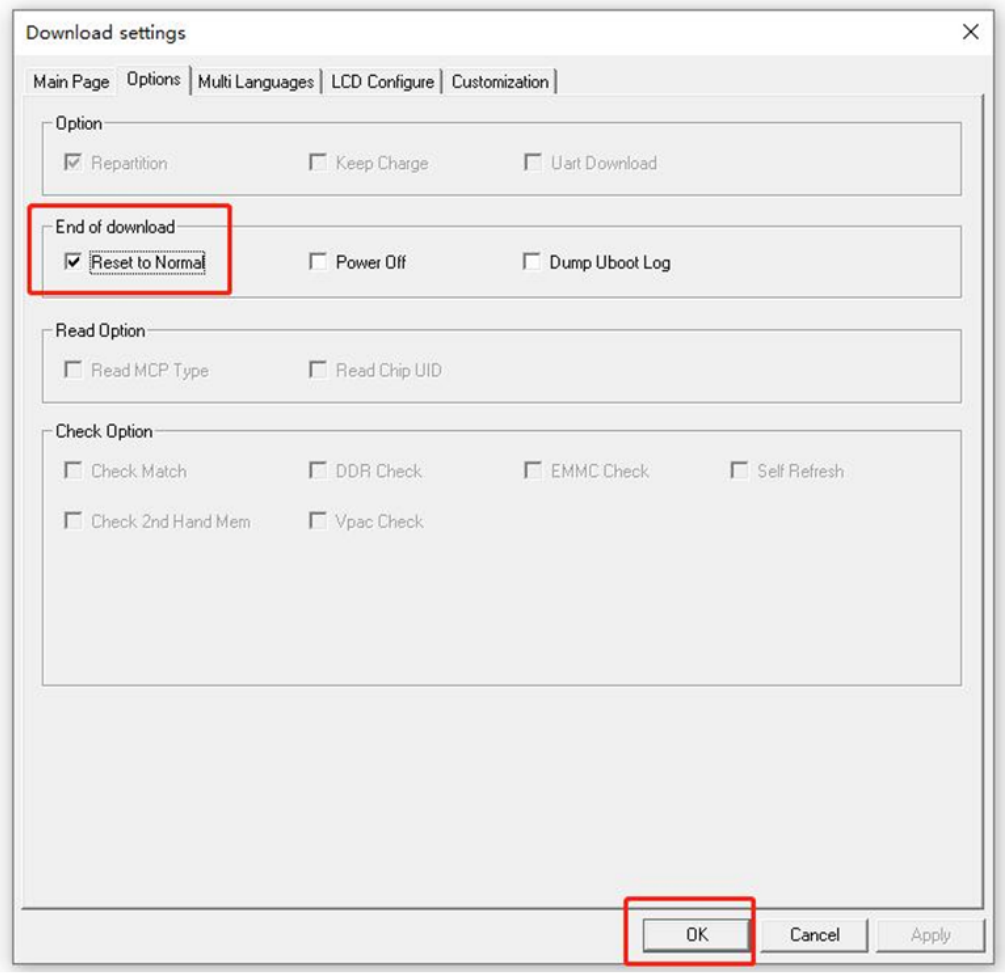

Figure 7: Whether to Automatically Reboot Module After Firmware Download

2. As shown in ***Figure 5***, click firmware pac file loading button (Button 1) to select the firmware pac file to be downloaded in the target directory. After loading, the basic information of the firmware package will be displayed in the middle of the UpgradeDownload toolbar.

3. Click the firmware download button (Button 3), and if the module is in the download mode, the firmware will be downloaded. As shown in the figure below, **Port 49** is in the download status.

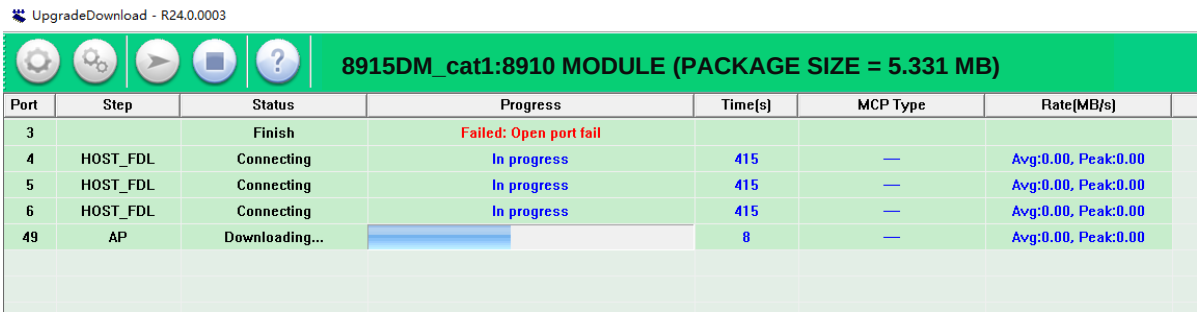

Figure 8: Firmware Download Status

If the module is not in download mode, "**Failed: Open port fail**" will be displayed after you click the firmware download button (Button 3). Then you need to put the module into download mode according to ***Chapter 2.2***.

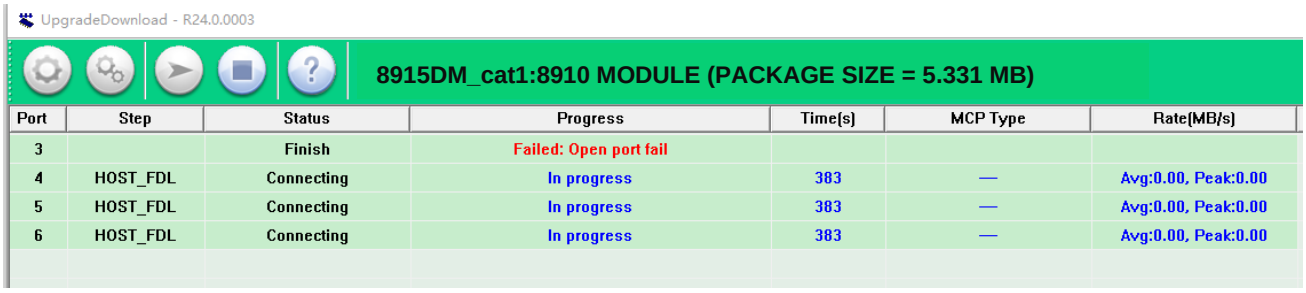

Figure 9: Opening the Firmware Download Port Failed

4. After downloading the firmware, the tool display is shown as the following figure:

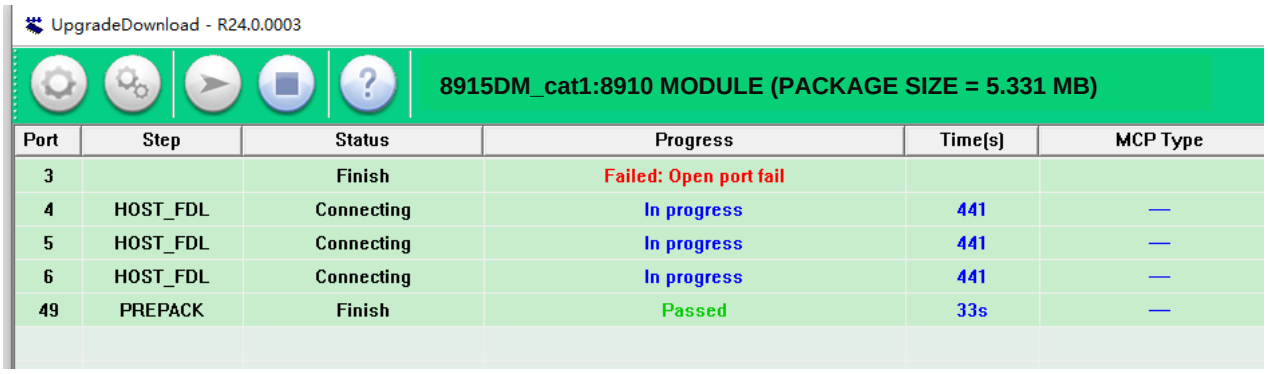

Figure 10: Firmware Download Completed

If it is configured to automatically reboot the module after the firmware is downloaded, it is shown as the figure below:

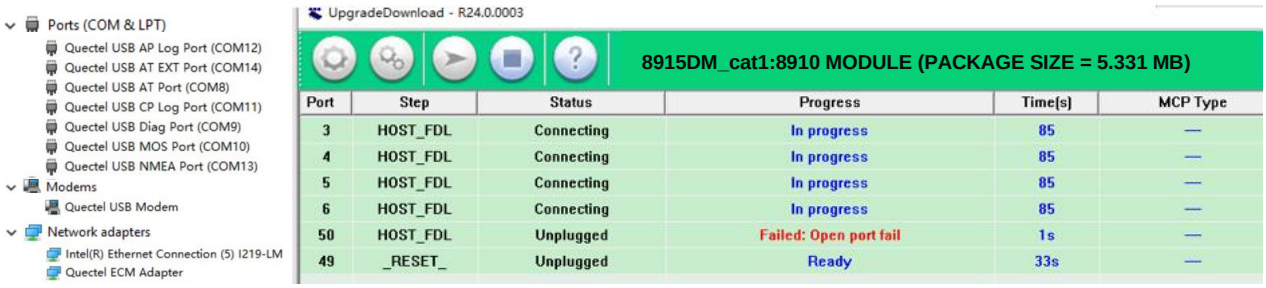

Figure 11: Reboot the Module Automatically After the Firmware Is Downloaded

3 Appendix References

Table 2: Terms and Abbreviations

Abbreviation	Description
APP	Application
IoT	Internet of Things
SDK	Software Development Kit
USB	Universal Serial Bus